AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT395
Applied Cybersecurity

Mid-term Examination MTA - Makeup
Fall Semester 2025/2026

Date: 4/Dec. /2025

lumber of Exam Pages: 3	**Time Allowed:** 90 m

Instructions:

- Total Marks: 30
- Total Number of Questions:3
- Marks Distribution:
  - Question 1: 10 marks
  - Question 2: 10 marks
  - Question 3: 10 marks
- Write your answer in the answer booklets. Answers given on the exam paper will not be marked.
- The use of electronic devices that could have a memory is NOT permitted
- Be sure you write your name and ID on the External Answer booklet.

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Question 1: Select the best answer : **Total: 10 Marks**

1. Which of the following usually observe each activity on the internet of the victim, gather all information in the background, and send it to someone else?
   a) Malware
   b) Spyware
   c) Adware
   d) All of the above
2. Which of the following ensures that only authorized parties can view protected information? (1 Mark)
   a. Authorization
   b. Confidentiality
   c. Availability
   d. Integrity
3. identifying users' needs. Which of these generally recognized security positions has Ian been offered?(1 Mark)
   a. Security administrator
   b. Security technician
   c. Security officer
   d. Security manager
4. A(n) _____ is a program that secretly attaches itself to a carrier such as a document or program and then executes when that document is opened or program is launched. (1 Mark)
   a. virus
   b. worm
   c. rootkit
   d. Trojan
5. The first action that a virus takes once it infects a computer is to(1 Mark)
   a. close all ports
   b. erase the payload
   c. authenticate
   d. replicate
6. A(n) _____ works on the principle of comparing new behavior against normal behavior.
   a. Host Intrusion Detection System (HIDS) (1 Mark)
   b. packet filter
   c. Internet Resource Chat (IRC)
   d. personal software firewall
7. A computer program that pretends to clean up a hard drive but actually performs a malicious activity is known as a(n) (1 Mark)
   a. Trojan
   b. Rootkit
   c. worm
   d. logic bomb
8. A(n) _____ is either a small hardware device or a program that monitors each keystroke a user types on the computer's keyboard. (1 Mark)
   a. keylogger
   b. macro
   c. script kiddie
   d. port replicator

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9. The _____ is the software that runs on a physical computer and manages multiple virtual machine operating systems. (1 Mark)
    a. virtual resource allocator (VRA)
    b. hypervisor
    c. Microsoft Control Plug-in
    d. hardware allocator
10. _____ is exploiting a vulnerability in the software to gain access to resources that the user would normally be restricted from obtaining. (1 Mark)
    a. Vulnerability assessment
    b. Software threat release (STR)
    c. Hardware virus
    d. Privilege escalation

Question 2: Describe the following general kinds of attacks, in a few sentences: Total: 10 Marks

1. Explain the difference between a worm and virus. (2 marks)
2. Explain the difference between a normal (parasitic) virus, a metamorphic virus and a polymorphic virus. (2 marks)
3. SYN flooding attack (2 marks)
4. SQL Injection attack (4 Marks)
5. Cross Site Scripting (XSS) (2 marks)

Question 3 Answer the following questions **[10 Marks]**

1. Explain the difference between packet filters and application layer proxies. (2 Marks)
2. Compare between IDS and IPS. (2 Marks)
3. Explain the three types of identity authentication tasks? (2 Marks)
4. Explain the difference between Stateless packet filtering and Stateful packet filtering. (2 Marks)
5. What is SIEM? List three of SIEM's Key Capabilities? (2 Marks)

End of questions

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